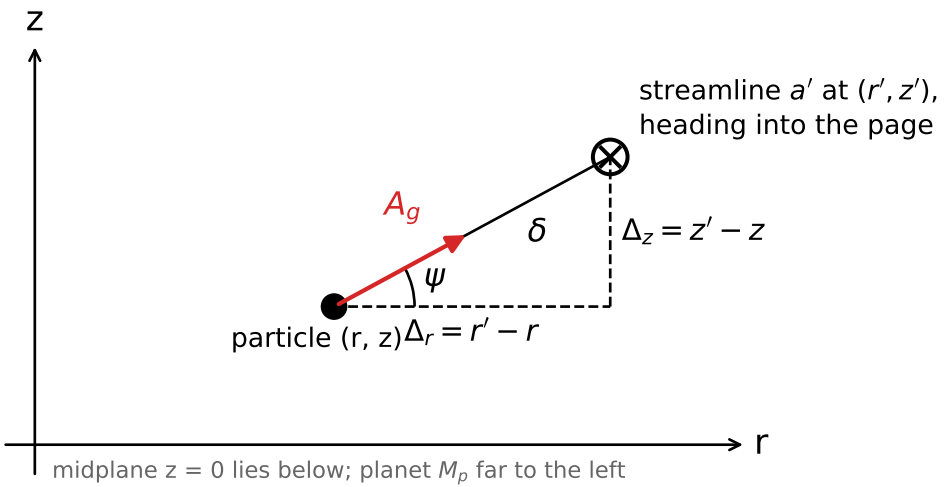
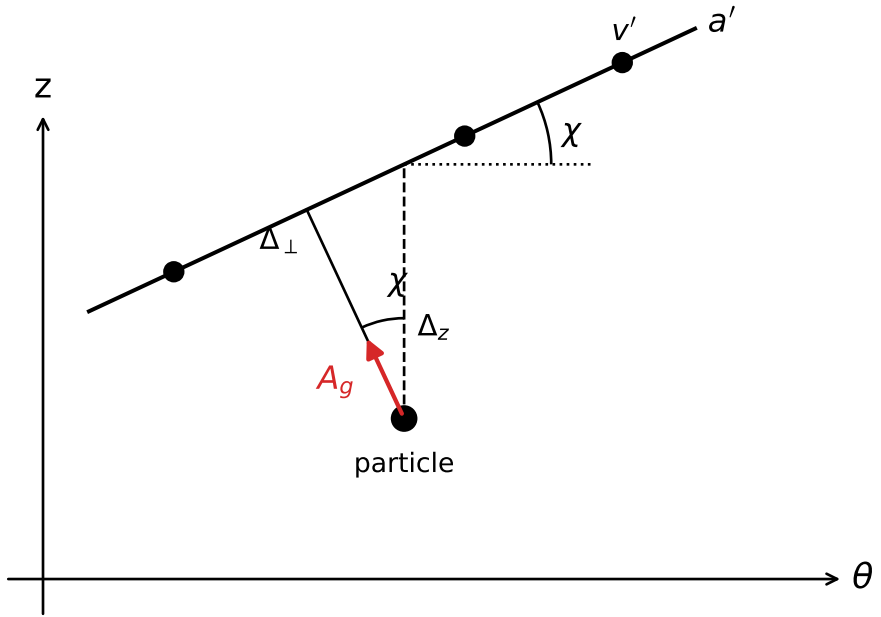
(a) meridional plane through the particle



(b) seen from outside the ring, along $-\hat{r}$